BIRATIONAL MORPHISMS BETWEEN SMOOTH THREEFOLDS MAY NOT DECOMPOSE INTO SMOOTH BLOW-DOWNS AND FLOPS

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Abstract. We give a negative answer to a question of Kollár: there exists a birational morphism between smooth projective threefolds which is not a composite of smooth blow-downs and flops.

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Any birational morphism between smooth projective surfaces is a composite of blow-downs to points [Kol08, Theorem 1]. In dimension three, smooth blow-downs are no longer sufficient. Let $C \subset \mathbb{P}^3$ be an irreducible curve whose only singular point is a node or a cusp, and let p be the unique singular point of $\mathrm{Bl}_C \mathbb{P}^3$. Then $\mathrm{Bl}_p \mathrm{Bl}_C \mathbb{P}^3 \to \mathbb{P}^3$ is not a composite of smooth blow-downs, but it is a composite of two smooth blow-downs and a flop [Kol08, Exercise 22]. Kollár asked whether any birational morphism between smooth projective threefolds is a composite of smooth blow-downs and flops [Kol08, Question 4].

The answer is known to be positive in several cases. Jiang settled the case when the relative Picard number is two [Jia25, Theorem 1.2]. Xie settled the case of projective toric morphisms between smooth quasi-projective toric threefolds, with toric flops as the flops [Xie26, Theorem 1.1], and gave an example showing that the toric statement fails without projectivity [Xie26, Example 5.1]. Chen proved a weaker factorization, in which flops are replaced by P-desingularizations of terminal $\mathbb{Q}$-factorial flops [Che24, Theorem 1.2]. In this paper, we show that the answer to Kollár's question is negative in general.

Theorem 1.1. *There exists a birational morphism $f\colon X \to Y$ between smooth projective threefolds such that X admits no flop and no smooth blow-down. In particular, f is not a composite of smooth blow-downs and flops.*

Theorem 1.1 disproves [Kol08, Question 4]. The proof shows that X admits no flop even if the base of the flopping contraction is not required to be projective. See Lemma 4.2 for details.

Sketch of the proof. We take Y to be a product of three elliptic curves, so that any rational curve on X is contracted by f. Inside Y we take an irreducible curve C whose singular points are, analytically locally, the monomial curve $t \to (t^3, t^5, t^7)$, which is not a local complete intersection. The blow-up of C has a copy of $\mathbb{P}^2$ over each singular point of C, and each such $\mathbb{P}^2$ contains exactly one singular point of the blow-up. We choose a small modification that keeps these planes and moves the singular points to the other end of the flopping curves, and then blow up the new singular points. On the threefold X obtained in this way, the canonical degree of any rational curve is non-zero, so X admits no flop. The exceptional divisors of f are planes with normal bundle $\mathcal{O}(-2)$, quadric cones, and one divisor over C whose fiber over each singular point of C contains two distinct intersecting curves, so X admits no smooth blow-down.

Structure of the paper. In Section 2 we recall the definitions and a consequence of the cone theorem. In Section 3 we construct f and compute the local structure of f over the singular points of C. In Section 4 we prove Theorem 1.1.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

Definition 2.1. Let X and X' be smooth threefolds. A *smooth blow-down* is a morphism $b\colon X \to X'$ which is the blow-up of X' along a closed point or along a smooth irreducible curve.

The exceptional divisor of a smooth blow-down is isomorphic to $\mathbb{P}^2$ with normal bundle $\mathcal{O}_{\mathbb{P}^2}(-1)$, or is a $\mathbb{P}^1$-bundle over a smooth curve. In particular, it is smooth.

Definition 2.2 ([Kol08, Definition 73 and Warning 74(6)])**.** Let X be a normal variety. A *flop* of X is a birational map $\phi\colon X \dashrightarrow X^+$ to a normal variety X^+ for which there exist a normal variety Z, proper birational morphisms $g\colon X \to Z$ and $g^+\colon X^+ \to Z$ such that

$$\dim \operatorname{Exc}(g), \dim \operatorname{Exc}(g^+) \leq \dim X - 2,$$

and an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor H on X, such that $g = g^+ \circ \phi$, $-H$ is g-ample, $\phi_* H$ is $\mathbb{R}$-Cartier and g^+-ample, and $K_X \equiv_Z 0$. We say that X *admits no flop* if any flop of X is an isomorphism.

Definition 2.3. Let $f\colon X \to Y$ be a birational morphism between smooth projective threefolds that is not an isomorphism. We say that f is a *composite of smooth blow-downs and flops* if f can be decomposed into a sequence of birational maps

$$X =: X_0 \dashrightarrow X_1 \dashrightarrow X_2 \dashrightarrow \cdots \dashrightarrow X_n := Y$$

such that each birational map $\phi_i : X_{i-1} \dashrightarrow X_i$ is either a smooth blow-down or a flop.

We use the following consequence of the cone theorem.

Lemma 2.4. *Let X be a smooth projective variety, and let $g\colon X \to Z$ be a proper birational morphism to a normal variety such that $K_X \equiv_Z 0$ and g is not an isomorphism. Then there exists a rational curve $R \subset X$ such that $g(R)$ is a point and $K_X \cdot R = 0$.*

Proof. Since X is projective and Z is separated, g is projective [Har77, Chapter II, Exercises 4.8(e) and 4.9]. Since g is not an isomorphism and Z is normal, g has a positive-dimensional fiber over some point $z \in Z$ by Zariski's main theorem. Possibly shrinking Z to an affine neighborhood of z, we may assume that Z is affine. Since g is birational and Z is affine, there exists $D \geq 0$ on X such that D is anti-ample/Z. Pick $0 < \epsilon \ll 1$, then $(X, \epsilon D)$ is klt and g is $(K_X + \epsilon D)$-negative, so there exists a $(K_X + \epsilon D)$-negative extremal ray/Z Σ. By the cone theorem, Σ is spanned by a rational curve R. R satisfies our requirement. □

3. The construction

3.1. **The curve.** For any $c \in \mathbb{C}\backslash\{0\}$, let A_c be the smooth projective curve which contains the affine curve $\{w^2 = s^4 - c\}$ as an open subset, and let $\pi_c\colon A_c \to \mathbb{P}^1$ be the double cover given by s. Since $c \neq 0$, the polynomial $s^4 - c$ has four distinct roots and does not vanish at $s = 0$. As $s^4 - c$ has even degree, π_c is unramified over $s = 0$ and $s = \infty$ and has four simple branch points, so A_c is an elliptic curve by the Riemann–Hurwitz formula. Let $A_i := A_{c_i}$ and $\pi_i := \pi_{c_i}$ for $(c_1, c_2, c_3) = (2, 3, 5)$, and let

$$Y := A_1 \times A_2 \times A_3, \quad T := (\mathbb{P}^1)^3, \quad \pi := \pi_1 \times \pi_2 \times \pi_3\colon Y \to T.$$

Then Y is an abelian threefold. In particular, Y contains no rational curve.

Let s_1, s_2, s_3 be the coordinates on the three factors of T, let $C_0 \subset T$ be the image of the morphism $\mathbb{P}^1 \to T$ given by $t \to (t^3, t^5, t^7)$, and let $C := \pi^{-1}(C_0)$ be the scheme-theoretic inverse image. Let $\mathfrak{q} := (y^2 - xz, yz - x^4, z^2 - x^3y) \subset \mathbb{C}[x, y, z]$.

Lemma 3.1. *Notation and conditions as above. The following hold.*

(1) *The ideal $\mathfrak{q}$ is the ideal of the image of $t \to (t^3, t^5, t^7)$ in $\mathbb{A}^3$.*
(2) *The curve C is integral, and the singular locus of C consists of sixteen points $p_1, \dots, p_{16}$.*
(3) *For any i, there are local analytic coordinates x, y, z on Y at p_i in which C is defined by $\mathfrak{q}$.*

Proof. (1) Let $\mathfrak{p}$ be the ideal of the image. Then $\mathfrak{q} \subset \mathfrak{p}$. Modulo $\mathfrak{q}$, any monomial is congruent to one of x^a, $x^a y$, $x^a z$ with $a \geq 0$, as replacing y^2, yz, z^2 by xz, x^4, x^3y lowers the total degree in y and z. Under $t \to (t^3, t^5, t^7)$, these monomials go to t^{3a}, t^{3a+5}, t^{3a+7}. This implies (1).

(3) The morphism $\mathbb{P}^1 \to T$ is injective on $\mathbb{P}^1 \setminus \{0, \infty\}$, as $u^3 = u^5 = 1$ implies $u = 1$, and the differential of this morphism is non-zero there. Near $t = 0$, C_0 is defined by $\mathfrak{q}$ in the coordinates (s_1, s_2, s_3), and near $t = \infty$, C_0 is defined by $\mathfrak{q}$ in the coordinates $(s_1^{-1}, s_2^{-1}, s_3^{-1})$, by (1) applied to the parameter t^{-1}. Each π_i is étale over $s_i = 0$ and $s_i = \infty$, so π is étale over the two points $(0, 0, 0)$ and (∞, ∞, ∞), and each of them has 8 preimages. These are the 16 points $p_1, \dots, p_{16}$, and (3) follows.

(2) We are left to show that C is integral. Over $\mathbb{C}(t)$, the function field of C is obtained by adjoining square roots of

$$f_1 := t^{12} - 2, \quad f_2 := t^{20} - 3, \quad f_3 := t^{28} - 5.$$

Each f_j has only simple roots and f_j have pairwise no common roots. Thus any non-empty product of the f_j has a simple root, and so it is not a square in $\mathbb{C}(t)$. We have that $\mathbb{C}(t)[w_1, w_2, w_3]/(w_j^2 - f_j)$ is a field of degree 8 over $\mathbb{C}(t)$. Since π is finite and flat, any irreducible component of C dominates C_0, so C is irreducible and generically reduced.

For any $t_0 \neq 0, \infty$, at most one f_j vanishes at t_0, and the vanishing order is ≤ 1. Over a neighborhood of the point (t_0^3, t_0^5, t_0^7), C is locally isomorphic to the curve $\{w_j^2 = f_j\}$, or is étale over C_0 if no f_j vanishes at t_0, so C is smooth there. At $p_1, \dots, p_{16}$, C is reduced and singular by (1) and (3). Thus C is reduced, hence integral, and the singular locus of C is $\{p_1, \dots, p_{16}\}$. □

3.2. **The blow-up of the curve.** Let $h\colon V := \mathrm{Bl}_C Y \to Y$ be the blow-up of C.

Lemma 3.2. *Let $V_0 := \mathrm{Bl}_{\mathfrak{q}} \mathbb{A}^3$ and let a, b, c be homogeneous coordinates on $\mathbb{P}^2$.*

(1) *$V_0 \subset \mathbb{A}^3 \times \mathbb{P}^2$ is defined by*

$$za - yb + xc = 0 \quad \text{and} \quad x^3a - zb + yc = 0. \tag{3.1}$$

(2) *The fiber of $V_0 \to \mathbb{A}^3$ over the origin is $D_0 \cong \mathbb{P}^2$, and V_0 is smooth outside the point $q := (0, [1 : 0 : 0]) \in D_0$.*
(3) *In the chart $a = 1$, with $v := c - b^2$ and $\eta := y + xb$, the variety V_0 is the hypersurface*

$$v\eta + x(x^2 + b^3) = 0 \tag{3.2}$$

in the coordinates x, b, v, η, and $D_0 = (x = \eta = 0)$.
(4) *V_0 is normal and Gorenstein, and $\omega_{V_0}|_{D_0} \cong \mathcal{O}_{\mathbb{P}^2}(-1)$.*

Proof. Let $g_1 := y^2 - xz$, $g_2 := yz - x^4$ and $g_3 := z^2 - x^3y$. Then $zg_1 - yg_2 + xg_3 = 0$ and $x^3g_1 - zg_2 + yg_3 = 0$, so V_0 is contained in the subscheme W of $\mathbb{A}^3 \times \mathbb{P}^2$ defined by (3.1). The 2×2 minors of the coefficient matrix of (3.1) are $-g_3$, g_2 and $-g_1$, and the first row $(z, -y, x)$ of this matrix is non-zero outside the origin. Thus outside the curve $\{\mathfrak{q} = 0\}$, W consists of the points $(u, [g_1(u) : g_2(u) : g_3(u)])$, and over a point of the curve other than the origin, the coefficient matrix has rank one, so the fiber of W is a line. Over the origin, both equations vanish identically, so the fiber is $\mathbb{P}^2$. Thus the part of W over the curve is of dimension two, and since W is defined by two equations in the smooth fivefold $\mathbb{A}^3 \times \mathbb{P}^2$, any irreducible component of W is of dimension at least 3 in $\mathbb{A}^3 \times \mathbb{P}^2$ and dominates $\mathbb{A}^3$. Hence W is irreducible and is a

complete intersection, so it is Cohen–Macaulay. Since W is generically reduced, it is reduced. Therefore $W = V_0$, which proves (1), and the fiber over the origin is $D_0 \cong \mathbb{P}^2$.

Outside the origin of $\mathbb{A}^3$, the curve is smooth, so V_0 is smooth there. At a point $(0, [a : b : c]) \in D_0$, the partial derivatives of (3.1) with respect to (x, y, z) are $(c, -b, a)$ and $(0, c, -b)$, and those with respect to (a, b, c) vanish. These two vectors are linearly independent unless $b = c = 0$, so V_0 is smooth outside q. This proves (2).

In the chart $a = 1$, the first equation of (3.1) is equivalent to $z = yb - xc$, and substituting this into the second equation, we obtain $x^3 + xbc + y(c - b^2) = 0$. Substituting $c = v + b^2$ and $y = \eta - xb$, we obtain (3.2). On D_0 we have $x = y = z = 0$, which is equivalent to $x = \eta = 0$. This proves (3).

Since V_0 is a complete intersection with an isolated singular point, it is normal and Gorenstein. Let $\mathrm{pr}_2 \colon \mathbb{A}^3 \times \mathbb{P}^2 \to \mathbb{P}^2$ be the projection. Both equations of (3.1) are sections of $\mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^2}(1)$, so by adjunction, ω_{V_0} is the restriction of $\omega_{\mathbb{A}^3 \times \mathbb{P}^2} \otimes \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^2}(2) \cong \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^2}(-1)$, and $\omega_{V_0}|_{D_0} \cong \mathcal{O}_{\mathbb{P}^2}(-1)$. This proves (4). □

By Lemmas 3.1(3) and 3.2, for any i, a neighborhood of $h^{-1}(p_i)$ in V is analytically isomorphic to a neighborhood of D_0 in V_0. We let $D_i := h^{-1}(p_i) \cong \mathbb{P}^2$ and let $q_i \in D_i$ be the point corresponding to q. Then V is a normal Gorenstein projective threefold, V is smooth outside $\{q_1, \dots, q_{16}\}$, $\omega_V|_{D_i} \cong \mathcal{O}_{\mathbb{P}^2}(-1)$ for any i, and the h-exceptional prime divisors are $D_1, \dots, D_{16}$ and the closure F_V of $h^{-1}(C \setminus \{p_1, \dots, p_{16}\})$ in V.

3.3. **The small modification.** Let $D := \sum_{i=1}^{16} D_i$, which is a Weil divisor on V, and let

$$p \colon U := \mathrm{Proj}_V \bigoplus_{n \geq 0} \mathcal{O}_V(D)^n \to V,$$

where $\mathcal{O}_V(D)^n$ is the image of $\mathcal{O}_V(D)^{\otimes n}$ in the function field of V. Thus p is the blow-up of the coherent fractional ideal $\mathcal{O}_V(D)$, so p is projective, and U is projective as V is projective.

Lemma 3.3. *Notation and conditions as above. The following hold.*

(1) *p is an isomorphism over $V \setminus \{q_1, \dots, q_{16}\}$.*
(2) *For any i, $p^{-1}(q_i) = \ell_i \cong \mathbb{P}^1$, and a neighborhood of ℓ_i in U is analytically isomorphic to a neighborhood of the exceptional curve of the blow-up of the hypersurface (3.2) along the ideal (x, v), which is covered by two charts: the affine space with coordinates v, α, b, on which*

$$x = v\alpha, \quad \eta = -\alpha(\alpha^2 v^2 + b^3), \tag{3.3}$$

and the hypersurface

$$\beta\eta + x^2 + b^3 = 0 \tag{3.4}$$

in the coordinates x, b, β, η, on which $v = x\beta$. On ℓ_i we have $\alpha\beta = 1$.
(3) *U has exactly sixteen singular points $q_1', \dots, q_{16}'$, where $q_i' \in \ell_i$ is the origin of (3.4).*
(4) *The strict transform of D_i is $(\alpha = 0)$ in (3.3). It is isomorphic to $D_i \cong \mathbb{P}^2$, intersects ℓ_i at the point $\alpha = 0$, and does not contain q_i'.*
(5) *$K_U = p^* K_V$. In particular, K_U restricts to $\mathcal{O}_{\mathbb{P}^2}(-1)$ on the strict transform of D_i, and $K_U \cdot \ell_i = 0$.*

Proof. D is Cartier on the smooth locus of V, so (1) holds. Near q_i, use the coordinates of (3.2). The divisor of x is $D_i + D_i'$, where $D_i' := (x = v = 0)$. Since (x, v) is a prime ideal of height one in a normal ring, it is the ideal of D_i', so $\mathcal{O}_V(D_i) = x^{-1}(x, v)$ near q_i, and p is the blow-up of the ideal (x, v) near q_i. Near q_i, U is the strict transform of the hypersurface (3.2) in the blow-up of $\mathbb{A}^4$ along (x, v) [Har77, Chapter II, Corollary 7.15]. If we write $x = v\alpha$ in (3.2) and remove the factor v, then we obtain $\eta + \alpha(\alpha^2 v^2 + b^3) = 0$, which is (3.3). If we write $v = x\beta$ in (3.2) and remove the factor x, then we obtain (3.4). This proves (2). The chart (3.3) is smooth, and the only singular point of (3.4) is the origin, which implies (3). In (3.3), $x = \eta = 0$ holds on $(\alpha = 0)$,

and $(\alpha = 0)$ maps isomorphically to D_i with coordinates v, b. The origin of (3.4) has $\beta = 0$, so it is not on $(\alpha = 0)$, which proves (4). Since p is small and U and V are normal and Gorenstein, $K_U = p^*K_V$, and (5) follows from (4) and $\omega_V|_{D_i} \cong \mathcal{O}_{\mathbb{P}^2}(-1)$. □

3.4. **The resolution.** Let $r\colon X \to U$ be the blow-up of $q'_1, \dots, q'_{16}$, let $S_i := r^{-1}(q'_i)$, and let $f := h \circ p \circ r\colon X \to Y$. By abusing notation, we still denote by D_i the strict transform of D_i on X for each i, and we let Γ_i be the strict transform of ℓ_i on X.

Lemma 3.4. *The following hold.*

(1) *X is a smooth projective threefold and f is birational.*
(2) *S_i is the quadric cone $\{\beta\eta + x^2 = 0\} \subset \mathbb{P}^3$, $K_X = r^*K_U + \sum_{i=1}^{16} S_i$, and $K_X|_{S_i} \cong \mathcal{O}_{S_i}(-1)$.*
(3) *r is an isomorphism over a neighborhood of the strict transform of D_i on U. Moreover, $D_i \cong \mathbb{P}^2$, $D_i \cap S_j = \emptyset$ for any j, $K_X|_{D_i} \cong \mathcal{O}_{\mathbb{P}^2}(-1)$ and $\mathcal{O}_{D_i}(D_i) \cong \mathcal{O}_{\mathbb{P}^2}(-2)$.*
(4) *$\Gamma_i \cong \mathbb{P}^1$, $S_i \cdot \Gamma_i = 1$, and $K_X \cdot \Gamma_i = 1$.*

Proof. We blow up the origin of (3.4), so near S_i, X is the strict transform of the hypersurface (3.4) in the blow-up of $\mathbb{A}^4$ at the origin [Har77, Chapter II, Corollary 7.15]. In the chart $(x, \beta, \eta) = b(x_1, \beta_1, \eta_1)$, the strict transform of the hypersurface (3.4) is $\beta_1\eta_1 + x_1^2 + b = 0$. In the chart $(x, b, \eta) = \beta(x_1, b_1, \eta_1)$, it is $\eta_1 + x_1^2 + \beta b_1^3 = 0$, and the chart $(x, b, \beta) = \eta(x_1, b_1, \beta_1)$ is symmetric. In the chart $(b, \beta, \eta) = x(b_1, \beta_1, \eta_1)$, it is $\beta_1\eta_1 + 1 + xb_1^3 = 0$, whose singular points would satisfy $\beta_1 = \eta_1 = b_1 = 0$, which contradicts the equation. Thus X is smooth, and it is projective as U is projective. Since h, p, r are birational, (1) holds.

The exceptional divisor S_i is the projectivized tangent cone $\{\beta\eta + x^2 = 0\} \subset \mathbb{P}^3$ with coordinates x, b, β, η, and $\mathcal{O}_{S_i}(S_i) \cong \mathcal{O}_{S_i}(-1)$. The blow-up of a point of a smooth fourfold has discrepancy 3, and a hypersurface with a double point acquires multiplicity 2 along the exceptional divisor, so $K_X = r^*K_U + \sum_i S_i$. Restricting to S_i, we obtain $K_X|_{S_i} \cong \mathcal{O}_{S_i}(S_i) \cong \mathcal{O}_{S_i}(-1)$. This proves (2).

By Lemma 3.3(4), D_i does not contain q'_j for any j, so r is an isomorphism near D_i. Thus $D_i \cong \mathbb{P}^2$, $D_i \cap S_j = \emptyset$, and $K_X|_{D_i} \cong K_U|_{D_i} \cong \mathcal{O}_{\mathbb{P}^2}(-1)$ by Lemma 3.3(5). By adjunction, $\mathcal{O}_{D_i}(D_i) \cong \omega_{\mathbb{P}^2} \otimes K_X|_{D_i}^{-1} \cong \mathcal{O}_{\mathbb{P}^2}(-2)$. This proves (3).

Near q'_i, the curve ℓ_i is $(x = b = \eta = 0)$ with coordinate β. In the chart $(x, b, \eta) = \beta(x_1, b_1, \eta_1)$, Γ_i is $(x_1 = b_1 = \eta_1 = 0)$ and S_i is $(\beta = 0)$, so $S_i \cdot \Gamma_i = 1$. Since $K_U \cdot \ell_i = 0$ by Lemma 3.3(5), we have $K_X \cdot \Gamma_i = K_U \cdot \ell_i + S_i \cdot \Gamma_i = 1$ by (2). This proves (4). □

4. Proof of Theorem 1.1

Notation as in Section 3. Let $F \subset X$ be the strict transform of F_V.

Lemma 4.1. *The following hold.*

(1) *The f-exceptional prime divisors are F, $D_1, \dots, D_{16}$ and $S_1, \dots, S_{16}$.*
(2) *For any i, $f^{-1}(p_i) = D_i \cup S_i \cup \Gamma_i$ as sets. For any $y \in C \setminus \{p_1, \dots, p_{16}\}$, $f^{-1}(y)$ is a smooth rational curve Σ with $K_X \cdot \Sigma = -1$, and f is an isomorphism over $Y \setminus C$.*
(3) *Any rational curve $R \subset X$ satisfies $K_X \cdot R \neq 0$.*

Proof. (1) and (2) follow from the construction: $h^{-1}(p_i) = D_i$, $p^{-1}(D_i)$ is the union of the strict transform of D_i and ℓ_i, r only modifies U at $q'_i \in \ell_i \setminus D_i$, and over $C \setminus \{p_1, \dots, p_{16}\}$, f is the blow-up of a smooth curve in a smooth threefold.

(3) Since Y contains no rational curve, $f(R)$ is a point, so R is contained in a fiber of f. If $R \subset D_i$, then $K_X \cdot R < 0$ by Lemma 3.4(3). If $R \subset S_i$, then $K_X \cdot R < 0$ by Lemma 3.4(2). If $R = \Gamma_i$, then $K_X \cdot R = 1$ by Lemma 3.4(4). Otherwise, R is a fiber of f over a point of $C \setminus \{p_1, \dots, p_{16}\}$, and $K_X \cdot R = -1$ by (2). □

Lemma 4.2. *X admits no flop.*

Proof. By Lemmas 2.4 and 4.1(3), any K_X-trivial birational morphism is an isomorphism. The lemma follows. □

Lemma 4.3. *For any i, there is a line $\Lambda_i \subset D_i$ such that $\Lambda_i \subset F$ and $\Lambda_i \cap \Gamma_i \neq \emptyset$. Moreover, $\Gamma_i \subset F$ and $\Gamma_i \neq \Lambda_i$.*

Proof. By Lemma 3.2(1), the fiber of $V_0 \to \mathbb{A}^3$ over the point (t^3, t^5, t^7) with $t \neq 0$ is the line $\{t^4a - t^2b + c = 0\} \subset \mathbb{P}^2$, as both equations of (3.1) become $t^3(t^4a - t^2b + c) = 0$ and $t^5(t^4a - t^2b + c) = 0$. For any $[a : b] \in \mathbb{P}^1$, the point $((t^3, t^5, t^7), [a : b : t^2b - t^4a])$ lies on this line and converges to $(0, [a : b : 0])$ as $t \to 0$. Thus the closure of the union of these lines contains the line $\{c = 0\} \subset D_0$, which passes through $q = [1 : 0 : 0]$. Let $\Lambda_i \subset D_i$ be the corresponding line. Then $\Lambda_i \subset F_V$ and $q_i \in \Lambda_i$. By Lemmas 3.3(4) and 3.4(3), the strict transform of Λ_i on X, still denoted by Λ_i, is contained in $D_i \cap F$ and passes through the point $D_i \cap \Gamma_i$, which is the point of D_i over q_i.

Since X and Y are smooth, we may write $K_{X/Y} = E \geq 0$ such that $\operatorname{Supp} E = \operatorname{Exc}(f)$. Since Γ_i is contained in a positive-dimensional fiber of f, $\Gamma_i \subset \operatorname{Supp} E$. By Lemmas 4.1(2), 3.4(3)(4), and 4.1(1), $\Gamma_i \subset F$. Since ℓ_i is not contained in the strict transform of D_i by Lemma 3.3(4), $\Gamma_i \not\subset D_i$, so $\Gamma_i \neq \Lambda_i$. □

Lemma 4.4. *X admits no smooth blow-down.*

Proof. Suppose that $b\colon X \to X'$ is a smooth blow-down with exceptional divisor E.

Step 1. In this step we show that f factors through b. Any fiber of b is a point, $\mathbb{P}^1$, or $\mathbb{P}^2$, and any morphism from these varieties to Y is constant since Y contains no rational curve. Let $G \subset X' \times Y$ be the image of $(b, f)\colon X \to X' \times Y$. The projection $G \to X'$ is proper and birational, and any fiber of $G \to X'$ is a point. Thus it is finite and birational, so it is an isomorphism as X' is normal. Hence $f = h' \circ b$ for some morphism $h'\colon X' \to Y$.

Step 2. In this step we conclude the proof. By **Step 1**, E is f-exceptional, so E is one of F, D_i, S_i by Lemma 4.1(1). Since S_i is singular, $E \neq S_i$. Since $\mathcal{O}_{D_i}(D_i) \cong \mathcal{O}_{\mathbb{P}^2}(-2)$ and $\mathbb{P}^2$ admits no morphism with one-dimensional fibers onto a curve, $E \neq D_i$. Suppose that $E = F$. Since $f(F) = C$ is a curve, b is the blow-up of a smooth irreducible curve $B \subset X'$, and F is a $\mathbb{P}^1$-bundle over B via b. Since $f|_F = h'|_B \circ b|_F$ and $h'|_B\colon B \to C$ is surjective, the fiber of $f|_F$ over p_i is, as a set, a disjoint union of fibers of $F \to B$, each of which is irreducible. This contradicts Lemma 4.3, as Λ_i and Γ_i are distinct irreducible curves in this fiber with $\Lambda_i \cap \Gamma_i \neq \emptyset$. □

Proof of Theorem 1.1. By Lemma 3.4(1), $f\colon X \to Y$ is a birational morphism between smooth projective threefolds, and it is not an isomorphism as C is a curve. By Lemmas 4.2 and 4.4, X admits no flop and no smooth blow-down. The theorem follows. □

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